Automatic Subtitling for Social Media Videos

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1 Introduction

In the era of social media, video content has become a dominant medium for communication. However, many viewers watch videos without sound, making subtitles essential for engagement. To address this, I developed an algorithm that automatically generates and overlays subtitles on videos, tailored specifically for social media platforms. This tool is designed to be cost-effective, customizable, and easy to use, offering a free alternative to premium services like Submagic, which charges $50 per month.

2 Technical Setup

To implement this solution, I utilized a combination of Python-based tools and libraries. The setup process involved the following steps:

2.1 Environment Setup

- **Anaconda**: I used Anaconda to manage Python environments and dependencies, ensuring a seamless integration of the required libraries.
- **Git**: Git was installed to clone the `auto-subtitle` repository from GitHub, which serves as the foundation for this project.
- **FFmpeg**: Instead of installing FFmpeg via package managers (e.g., `choco install ffmpeg`), I opted to install it via `pip`, simplifying the setup process and avoiding additional installations.

2.2 Installation

The installation process involved the following commands:

```
pip install git+https://github.com/m1guelpf/auto-subtitle.git
pip install ffmpeg
```

3 Algorithm Overview

The algorithm leverages OpenAI's Whisper for speech-to-text transcription and FFmpeg for video processing. The key features and modifications I implemented include:

3.1 Automatic Subtitling

- The tool automatically generates subtitles for any video using Whisper's transcription capabilities.
- Subtitles are overlaid on the video, making it accessible to viewers who watch without sound.

3.2 Customization for Social Media

- **Captions at the Center**: Instead of traditional subtitles at the bottom, I modified the script to position captions at the center of the video, making them more visually appealing for social media platforms.
- **Word Highlighting**: To enhance engagement, I implemented a feature that changes the color of words as they are spoken, creating a dynamic and interactive viewing experience.
- **ASS Files**: Instead of using SRT files, I utilized ASS (Advanced SubStation Alpha) files for greater flexibility in styling and positioning subtitles.

3.3 Whisper Integration

- I used Whisper's word-level timestamps to break down sentences into smaller groups of words, ensuring precise synchronization between the audio and subtitles.
- The ability to choose different Whisper models (e.g., `small`, `medium`, `large`) allows for improved accuracy, especially for non-English content.

4 Usage

The tool is designed to be user-friendly and accessible. Key usage scenarios include:

4.1 Basic Usage

To generate subtitles for a video, the following command is used:

```
auto_subtitle /path/to/video.mp4 -o subtitled/
```

4.2 Advanced Options

- **Model Selection**: Users can specify the Whisper model for transcription. For example:

```
auto_subtitle /path/to/video.mp4 --model medium
```

- **Translation**: Subtitles can be translated into English using the `--task translate` option:

```
auto_subtitle /path/to/video.mp4 --task translate
```

5 Applications

This tool is particularly useful for content creators, influencers, and marketers who want to enhance their social media videos with professional-looking captions. As a practical example, I plan to use this tool to create engaging content for my stamp collection channel, showcasing my collection while making the videos accessible to a wider audience.

6 Conclusion

By combining OpenAI's Whisper, FFmpeg, and custom Python scripts, I developed a powerful and cost-effective solution for automatic subtitling. This tool not only saves time and money but also provides a high level of customization, making it an ideal choice for social media content creators. With its user-friendly interface and advanced features, it stands as a free and open-source alternative to premium services like Submagic.